

levels in GPT-4 responses and between human and machine evaluations, when aggregated to the task level. This exposure measure reflects an estimate of the technical capacity to make human labor more efficient; however, social, economic, regulatory, and other determinants imply that technical feasibility does not guarantee labor productivity or automation outcomes. Our analysis indicates that approximately 19% of jobs have at least 50% of their tasks exposed when considering both current model capabilities and anticipated tools built upon them. Human assessments suggest that only 3% of U.S. workers have over half of their tasks exposed to LLMs when considering existing language and code capabilities without additional software or modalities. Accounting for other generative models and complementary technologies, our human estimates indicate that up to 49% of workers could have half or more of their tasks exposed to LLMs.

Our findings consistently show across both human and GPT-4 annotations that most occupations exhibit some degree of exposure to LLMs, with varying exposure levels across different types of work. Occupations with higher wages generally present with higher exposure, a result contrary to similar evaluations of overall exposure to machine learning (Brynjolfsson et al., 2023). When regressing exposure measures on skillsets using O*NET's skill rubric, we discover that roles heavily reliant on science and critical thinking skills show a negative correlation with exposure, while programming and writing skills are positively associated with LLM exposure. Following Autor et al. (2022a), we examine barriers to entry by "Job Zones" and find that occupational exposure to LLMs weakly increases with the difficulty of job preparation. In other words, workers facing higher (lower) barriers to entry in their jobs tend to experience more (less) exposure to LLMs.

We further compare our measurements to previous efforts documenting the distribution of automation exposure in the economy and find broadly consistent results. Most other technology exposure measures we examine are statistically significantly correlated with our preferred exposure measure, while measures of manual routineness and robotics exposure show negative correlations. The variance explained by these earlier efforts (Acemoglu and Autor, 2011a; Frey and Osborne, 2017; Brynjolfsson et al., 2018; Felten et al., 2018; Webb, 2020; Brynjolfsson et al., 2023), along with wage controls, ranges from 60 to 72%, indicating that 28 to 40% of the variation in our AI exposure measure remains unaccounted for by previous technology exposure measurements.

We analyze exposure by industry and discover that information processing industries (4-digit NAICS) exhibit high exposure, while manufacturing, agriculture, and mining demonstrate lower exposure. The connection between productivity growth in the past decade and overall LLM exposure appears weak, suggesting a potential optimistic case that future productivity gains from LLMs may not exacerbate possible cost disease effects (Baumol, 2012; Aghion et al., 2018).³

Our analysis indicates that the impacts of LLMs like GPT-4, are likely to be pervasive. While LLMs have consistently improved in capabilities over time, their growing economic effect is expected to persist and increase even if we halt the development of new capabilities today. We also find that the potential impact of LLMs expands significantly when we take into account the development of complementary technologies. Collectively, these characteristics imply that Generative Pre-trained Transformers (GPTs) are general-purpose technologies (GPTs).⁴ (Bresnahan and Trajtenberg, 1995; Lipsey et al., 2005).

(Goldfarb et al., 2023) argue that machine learning as a broad category is likely a general-purpose technology. Our evidence supports a wider impact, as even subsets of machine learning software meet the criteria for general-purpose technology status independently. This paper's primary contributions are to provide a set of measurements of LLM impact potential and to demonstrate the use case of applying LLMs to develop such measurements efficiently and at scale. Additionally, we showcase the general-purpose potential of LLMs. If "GPTs are GPTs," the eventual trajectory of LLM development and application may be challenging for

³Baumol's cost disease is a theory that explains why the cost of labor-intensive services, such as healthcare and education, increases over time. This happens because wages for skilled workers in other industries increase, but there is no corresponding increase in productivity or efficiency in these service industries. Therefore, the cost of labor in these industries becomes relatively more expensive compared to other goods and services in the economy.

⁴For the remainder of the paper we spell out general-purpose technologies when it is used outside of stating "GPTs are GPTs."

2020; Benzell et al., 2021). The realization of general purpose technologies' full potential requires extensive co-invention (Bresnahan and Trajtenberg, 1995; Bresnahan et al., 1996, 2002; Lipsey et al., 2005; Dixon et al., 2021), a costly and time-consuming process involving the discovery of new business procedures (David, 1990; Bresnahan, 1999; Frey, 2019; Brynjolfsson et al., 2021; Feigenbaum and Gross, 2021). Consequently, many studies of machine learning technologies focus on systems-level adoption, arguing that organizational systems may require redesign to effectively take advantage of novel machine learning advancements (Bresnahan, 2019; Agrawal et al., 2021; Goldfarb et al., 2023). Appropriately designed systems can yield considerable business value and improve firm performance (Rock, 2019; Babina et al., 2021; Zolas et al., 2021), with AI tools facilitating the discovery process (Cockburn et al., 2018; Cheng et al., 2022). By employing task-level information to assess whether LLMs fulfill the criteria of a general purpose technology, we seek to merge the two perspectives for understanding the technology-labor relationship.

We attempt to build on these diverse literature streams in several ways. Echoing (Felten et al., 2023), we focus our analysis on the impact of LLMs, rather than addressing machine learning or automation technologies more broadly. Additionally, we propose a novel method that employs LLMs, specifically GPT-4, to assess tasks for exposure and automation potential, thereby bolstering human scoring efforts. Subsequently, we aggregate our findings to occupations and industries, capturing the overall potential exposure in the contemporary U.S. labor market.

3 Methods and Data Collection

3.1 Data on Activities and Tasks Performed by Occupation in the US

We use the O*NET 27.2 database (O*NET, 2023), which contains information on 1,016 occupations, including their respective Detailed Work Activities (DWAs) and tasks. A DWA is a comprehensive action that is part of completing task, such as "Study scripts to determine project requirements." A task, on the other hand, is an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. We offer a sample of tasks and DWAs in Table 1. The two datasets we use consist of:

- 19,265 tasks, consisting of a "task description" and a corresponding occupation, and
- 2,087 DWAs, where most DWAs are connected to one or more tasks, and tasks may be associated with one or more DWAs, though some tasks lack any associated DWAs.

3.2 Data on Wages, Employment, and Demographics

We obtain employment and wage data from the 2020 and 2021 Occupational Employment series provided by the Bureau of Labor Statistics. This dataset encompasses occupational titles, the number of workers in each occupation, and occupation-level employment projections for 2031, typical education required for entry in an occupation and on-the-job training required to attain competency in an occupation (BLS, 2022). We use the BLS-recommended crosswalk to O*NET (BLS, 2023b) to link the O*NET task and DWA dataset and the BLS Labor Force Demographics (BLS, 2023a), which is derived from the Current Population Survey (CPS). Both of these data sources are collected by the U.S. government and primarily capture workers who are not self-employed, are documented, and are working in the so-called formal economy.

3.3 Exposure

We present our results based on an exposure rubric, in which we define **exposure** as a measure of whether access to an LLM or LLM-powered system would reduce the time required for a human to perform a specific